SEA:ME 2026/27 · TEAM 1

ROS Rangers

Sprint 2 retrospective: protocol, gateway and failsafe, 2026-10-30 to 2026-11-13

2026-11-13 · BrunoBMFT · Dot99 · ptorrao-git · ThiagoTEP

SPRINT 2 · PLAN AGAINST RESULT

What we planned for weeks 3 and 4

Planned (Study Plan v2)	Result on 2026-11-13
Freeze Protocol Spec v0 in a design review	Done, 2026-11-03
C++ protocol library and Python twin, shared vectors	Done, 2026-11-05
Allow-listed gateway firmware on the ESP	Done, 2026-11-06
ESP link state machine, 250 ms failsafe	Done, 2026-11-10
flight_interface lifecycle node, watchdog, E-stop	Part 1 merged, part 2 draft (#187)
HIL job: flash from CI, failsafe tests	Done, 2026-11-13; HIL green again
Architecture document	Draft; waits on #103
Bench §11 Phases 1–5, MSP Override mask 8	Not run: waits on #123
Sprint 2 demo: fault injection on the bench	Not run: waits on #103
ThoneFlow parser and SoftwareSerial spike	Not started

SPRINT 2 · TIMELINE

Two weeks, day by day

Date	Milestone
2026-10-30	Safety Rule 18 (hands off an armed drone); Betaflight 2026.6.2 recorded
2026-11-03	Protocol Spec v0 accepted; HIL Wi-Fi window and first SoftAP tests
2026-11-04	Module 1 architecture document draft
2026-11-05	C++ library and Python twin; ADR-024 (OTA); one key per direction
2026-11-06	Gateway firmware with MSP allow-list; PX4 offboard bridge
2026-11-09	fc_telemetry, sd::Link state machine, flight_interface node, Pi as fake FC
2026-11-10	OTA gate and arming lock; HIL session tests; HIL finds a gateway stall
2026-11-12	Betaflight SITL TCP race fixed, CI stable again (#181)
2026-11-13	FC health and RC-ack rule (#183, #191); HIL green (#192); drone back, no USB

PROTOCOL · TRACKS B AND C

One spec, two languages, same test vectors

#97 #142 · 2026-11-03

Spec frozen

Protocol Spec v0 reviewed in one hour, the open items decided and recorded.

#143 #144 · 2026-11-05

C++ and Python

Encode, decode, HMAC and sequence window. Both pass the same golden vectors in CI.

#151 #155

Reflection closed

One key per direction, so a frame sent back to its sender is refused. Role and handshake rules enforced in code.

PROTOCOL · WHAT GOES OVER THE LINK

A session, message by message

Step	Messages	Link	What it does
Open a session	HELLO → HELLO_ACK	TCP	Signed with the shared key. Both sides add a random nonce and derive one key per direction. A replayed HELLO cannot end a live session
Take control	ARM_CONTROL_REQUEST → ARM_CONTROL_ACK	TCP	The ground states the override mask it expects; the drone refuses if its own differs
Fly	SETPOINT 50 Hz ↑ · STATE 25 Hz ↓	UDP	Roll, pitch, yaw rate, throttle up; attitude, battery, flow and link counters down. No gesture = neutral SETPOINT, never silence
Link lost	nothing arrives	–	150 ms: neutral sticks. 250 ms: LINK_FAILSAFE
Recover	RESUME_REQUEST → RESUME_ACK	TCP	Only after 500 ms of neutral SETPOINT. Never automatic
Tune, end	CONFIG_GET / SET → CONFIG_ACK · BYE	TCP	No CONFIG_SET while streaming; safety limits change only in firmware

PROTOCOL · WHAT THE DRONE REPORTS

Five states, four link events

BOOT → NO_SESSION → SESSION_IDLE → STREAMING → LINK_FAILSAFE

EVENT 0x0101

Setpoint stale: nothing for 150 ms, sticks go neutral and the drone levels

EVENT 0x0102

LINK_FAILSAFE entered: nothing for 250 ms

EVENT 0x0103

Resumed: the ground asked, after 500 ms of neutral sticks

EVENT 0x0104

Failsafe output still active when a new session starts

Every refused frame is counted with its reason (bad tag, old or repeated seq, wrong direction, wrong size…); STATE carries the total as rx_rejected.

ESP FIRMWARE · TRACK B

The ESP is now a gateway, not a bridge

Ground

ROS 2 nodes, signed frames

→

ESP gateway

MSP allow-list · session over TCP, control over UDP · 250 ms link failsafe · OTA gated, arming locked during upload

→

Flight controller

Betaflight 2026.6.2, MSP

#160 · 2026-11-06

FC link with the MSP allow-list; anything else is dropped

#170 #174 · 2026-11-09

sd::Link state machine as pure logic, then wired into the gateway

#177 · ADR-024

OTA only when allowed, no arming while an upload runs

#183 · 2026-11-13

FC health to the ground; ARM refused while the FC is silent

Merged 2026-11-13 #191: ARM and RESUME only after 1 s of good RC acks from the FC (#188)

GROUND AND SIMULATION · TRACKS C AND E

Telemetry in, setpoints out

#169 #175 · fc_telemetry

The FC talks to ROS 2

MSP attitude and battery published on /fc/attitude and /fc/battery.

#178 · flight_interface part 1

Lifecycle node

Activate and deactivate, a watchdog, and an E-stop that cuts setpoints. Unit tested.

Draft · decisions open

ESP backend, #187

Draft with DECISIONS.md: the team picks the open options on the PR, then the rest of #103 is written.

#154 · simulation

PX4 offboard bridge

Sim-only bridge and a one-command SITL bring-up. In review: make sim-smoke, the gesture chain in PX4 SITL (#190).

CI AND HIL · TRACK D

The HIL bench found a stall, and its cause

New HIL tests on the Pi runner, every merge to dev:

#140 #173

Wi-Fi window, SoftAP tests, the Pi as a fake FC

#180

BOOT, NO_SESSION and session tests; HIL PSK fixed

#192 · 2026-11-13

Gateway is the default build; one Wi-Fi window; #106 closed

2026-11-10 to 2026-11-13

9 of 10

HIL runs failed on dev after #177. The firmware was a suspect, but the same build connected 40 of 40 times from a laptop. The cause was the Pi's Wi-Fi power saving.

Power saving off in the SoftAP window (#192): HIL green on dev again. Not fully solid yet: with it off, one bench run of three still failed.

SAFETY · BRUNOBMFT

One bench incident this sprint

#184 · follow-up #189

FC USB torn off

Back from Tech3 on 2026-11-13 and working, but still without USB: the new connector arrives in about two weeks. No flight before the #189 checks.

It filed with the Incident form. Closed this sprint: #69, #126, #131, #184.

RETROSPECTIVE

Sprint 2 in numbers

36
merged pull requests

31
issues closed

25
new issues opened

24
accepted ADRs (+1: ADR-024)

1
bench incident

23 k
lines added on dev

RETROSPECTIVE · ALL · DRAFT, AGREE BEFORE THE TALK

What we keep, what we change

Went well

- Spec frozen, then built in C++ and Python in two days
- Review found the reflection flaw before any flight
- HIL caught a stall, then a bench A/B found its cause
- Every incident filed with the Incident form

Didn’t go well

- One bench incident: the FC USB torn off
- One drone, a week in repair, still no USB: demo and §11 bench slipped
- HIL red on dev for three days
- ThoneFlow and the architecture doc slipped

We change

- Short bench sessions, one goal, written checklist read out loud
- Strain relief and unplug rule (#184)
- No firmware merge while HIL is red

NEXT

Sprint 3: 16–27 Nov

CARRY OVER

Sprint 2 demo, fault injection (#109) · rest of flight_interface (#103, #187) · MSP budget spike (#123), then §11 bench, mask 8 (#104) · architecture document (#98)

16–20 NOV

Cascade controller in PX4 SITL behind flight_interface · flow, IMU and baro height estimate · real-time profile (#146) · props-off bench without USB

23–27 NOV

Position hold in simulation · gestures drive flight_interface in simulation · Sprint 3 demo on 2026-11-27

No flight until the new USB connector is in and #189 passes: USB and diff all check, failsafe re-test. Expected around 2026-11-27.

Questions?